1(c)(ii)	power output $= 22v = 22 \times 11 = 240\,W$
2(a)	Gravitational potential at a point is the work done per unit mass in bringing a small test mass from infinity to that point.

No.	Solution
2(b)	$\Delta E_p = E_{p,final} - E_{p,initial}$ $= \left(-\dfrac{GMm}{r_{final}}\right) - \left(-\dfrac{GMm}{r_{initial}}\right)$ $= -(6.67\times10^{-11})(6.0\times10^{24})(1600)\left(\dfrac{1}{2.7\times10^{7}} - \dfrac{1}{6400\times10^{3}}\right)$ $= 7.6\times10^{10}$ J
2(c)(i)	The gravitational force on the satellite provides for the centripetal force required for it to orbit around the planet. $F_G = F_C$ $\dfrac{GMm}{r^2} = \dfrac{mv^2}{r}$ where v is the orbital speed. $mv^2 = \dfrac{GMm}{r}$ $\therefore E_k = \dfrac{1}{2}mv^2 = \dfrac{GMm}{2r}$
2(c)(ii)	$E_k = \dfrac{GMm}{2r}$ $= \dfrac{(6.67\times10^{-11})(6.0\times10^{24})(1600)}{2(2.7\times10^{7})}$ $= 1.2\times10^{10}$ J
2(c)(iii)	The expression in **(c)(i)** only applies for a mass in circular orbit around the planet. When the satellite is on the surface of the Earth, it is not in orbit as the centripetal force on the satellite is due to the gravitational force and the normal contact force by the ground. Hence, the student's method in determining the initial kinetic energy is incorrect.
3(a)	